VerSe test set: model trained on TotalSegmentator vs model trained on VerSe

Table 1: Dice per vertebra on the VerSe test set (VerSe19 validation + test, 67 scans), mean / median, for the two models (layout of Table 2 of the original report). *Limit*: Dice of the ground truth after resampling to the model's voxel size and back, i.e. the highest score possible at that voxel size. *n*: number of scored vertebrae.

Trained on (voxel size)	L1	L2	L3	L4	L5	All	Limit	*n*
TotalSegmentator (902 scans) (1.5 mm)	.90 / .94	.91 / .94	.92 / .94	.91 / .94	.89 / .93	**.91 / .94**	.96	305
VerSe19 training set (56 scans) (1 mm)	.58 / .77	.72 / .94	.83 / .94	.83 / .94	.86 / .93	**.76 / .93**	1.00	305

Findings. Both models use the same network, training recipe and test set; only the training data differ (902 TotalSegmentator scans at 1.5 mm vs 56 VerSe19 training scans at 1 mm). On a typical scan both outline the vertebrae equally well (median Dice .93 vs .94), but the VerSe-trained model is clearly worse on average (mean .76 vs .91), and L1 suffers most. The loss comes from *naming* rather than outlining: 7 scans are counted one level off (vs 1), and the model labels 72 whole vertebrae as lumbar that are not (vs 26), mostly neck and thoracic vertebrae in the many VerSe scans that show the whole spine. With only 56 scans, all cropped to the lumbar region, it has not learned well enough what is *not* a lumbar vertebra. Here, many training scans from another source beat few scans from the target source.